AI COLLEGE ASSISTANT

Project Documentation

Project Type	AI-assisted student information chatbot
Primary Purpose	Answer common questions about subjects, notes, timetables, and college information
Knowledge Source	A defined, curated knowledge base
Suggested Level	Beginner-friendly Python project

Prepared as a project report for learning, documentation, and GitHub portfolio use.

Note: This document describes a proposed project design. Features and sample responses should be adapted to match the implementation you build.

1. Project Overview

The AI College Assistant is a student-support chatbot designed to answer frequently asked college-related questions from a defined knowledge base. Students can ask about subjects, study notes, class timetables, campus services, and other approved college information. The assistant searches the available information and returns a concise response.

The project can begin as a simple Python command-line application and later be extended into a web interface. To keep answers dependable, the assistant should answer from the information supplied to it and clearly say when a requested detail is not available.

2. Problem Statement

Students often need to check multiple documents or contact staff for routine information such as course topics, class timings, notes, room details, and college procedures. This can take time, especially when information is spread across several sources. A centralized assistant can make approved information easier to find.

3. Objectives

• Provide quick answers to common student questions.

• Organize subject details, notes links, timetables, and college FAQs in one knowledge base.

• Reduce repeated manual searching for routine information.

• Return understandable answers and guide students to official sources when appropriate.

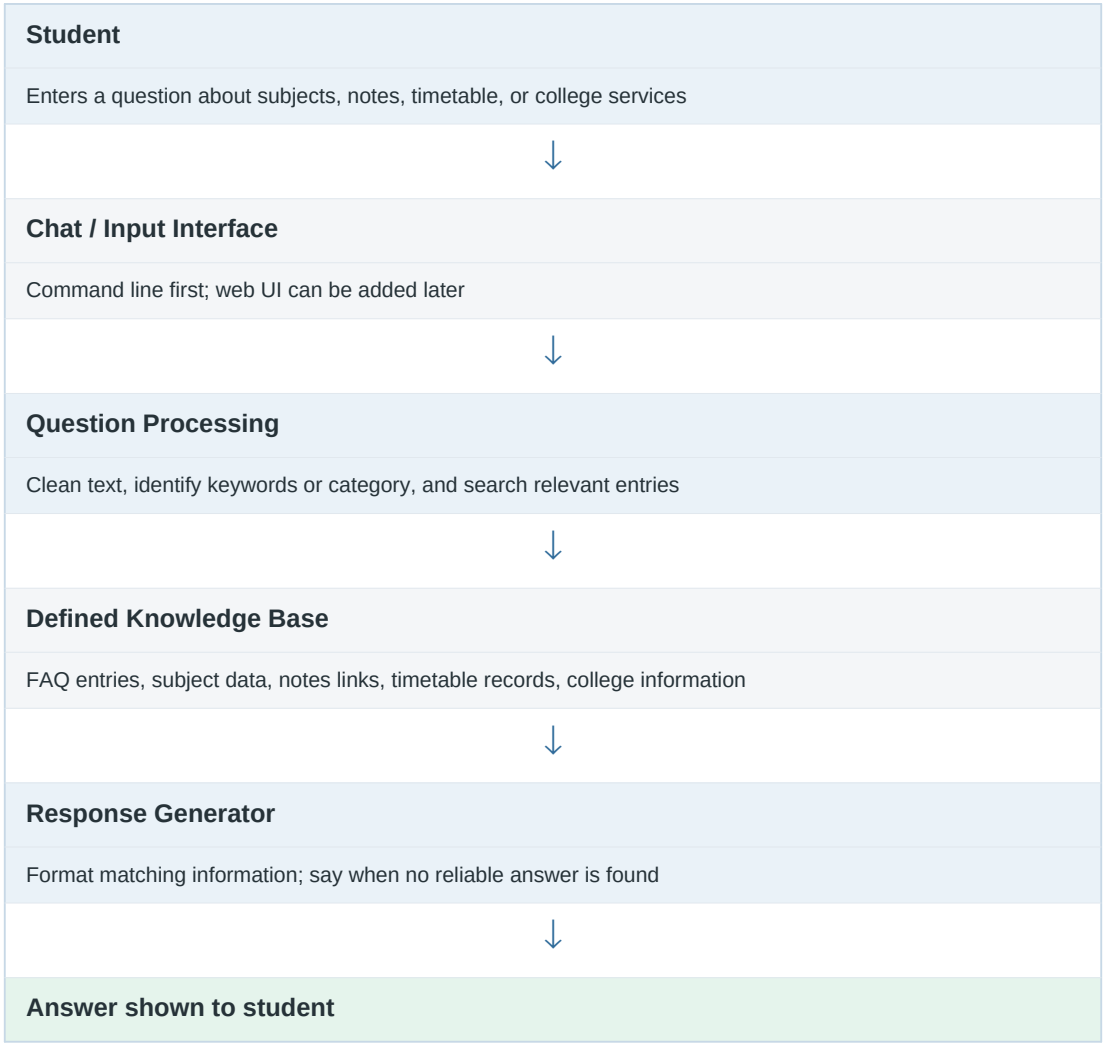
• Avoid inventing details when the knowledge base does not contain an answer.

4. Main Features

Feature	Description
Subject information	Course names, syllabus topics, subject codes, and faculty details when provided.
Notes and study resources	Find approved notes, document names, and links stored in the knowledge base.
Timetable lookup	Show class times, rooms, and days when accurate timetable data is available.
College FAQs	Answer questions about departments, facilities, office hours, and procedures from approved entries.
Unknown-answer handling	Tell the student when information is missing and suggest checking the official college source.

5. System Architecture

The architecture below shows a simple retrieval-based design. The student enters a question, the application normalizes it, searches the defined knowledge base, and formats the best matching information as a response.



6. Suggested Technology Stack

Component	Suggested choice	Purpose
Programming language	Python 3	Core logic and question handling
Knowledge base	JSON or CSV	Store FAQs and college data in a simple editable format
Search	Keyword matching; optional TF-IDF later	Find relevant entries for a question
Interface	Terminal initially; Flask/Streamlit later	Let students ask questions

7. Knowledge Base Design

Keep information structured so it can be updated without changing the main Python program. For a beginner version, a JSON file is a practical choice.

Example JSON record:

```
{ "category": "timetable", "question": "When is the first lecture?", "answer": "Check the latest official timetable for your section.", "source": "official_timetable.pdf", "last_updated": "YYYY-MM-DD" }
```

Recommended fields include category, question or keywords, answer, source, and last_updated. Replace the example with verified college information before using it.

8. Example User Interaction

Student question	Assistant response (example)
Where can I find notes for Mathematics?	The knowledge base lists the approved Mathematics notes resource. Open the stored link or ask your department if the resource is missing.
What is my class timetable?	Please provide your course and section, or check the latest official timetable. I can only show timetable data stored in the knowledge base.
What is the syllabus for a subject?	I can show the syllabus topics available in the knowledge base. Please enter the subject name or code.

9. Implementation Plan

1. Create a Python script that accepts a question from the user.
2. Prepare a JSON or CSV knowledge base with verified FAQs, subjects, timetable details, and resource links.
3. Normalize the question and search for the most relevant entry.
4. Display the matching answer and its source when available.
5. Add a clear fallback message for questions that cannot be answered.
6. Test with common questions, spelling variations, missing data, and invalid input.
7. Optionally add a web interface and administrative workflow for updating information.

10. Privacy, Accuracy, and Limitations

• Use approved, current college information and include sources or update dates where possible.
• Do not place private student records, passwords, marks, or personal contact details in a public repository.
• Restrict access if the project later handles non-public college information.
• Timetables and procedures may change; direct students to official sources for confirmation.
• A basic keyword-based assistant may misunderstand ambiguous questions and is not a substitute for official college staff.

11. Testing Checklist

■ Correct answers for known questions

■ Helpful fallback when information is missing

■ Works with empty input and extra spaces

■ Handles different capitalization and common wording

■ Sources and update dates are shown when available

■ No secrets or private student data committed to GitHub

12. Conclusion

The AI College Assistant provides a practical starting point for a student-support chatbot. By keeping its knowledge in a defined, maintainable data source, the project can offer quick access to common academic and campus information while making uncertainty clear. It can be expanded gradually with better search, a graphical interface, and an approved process for updating content.

Suggested repository files

README.md • main.py • knowledge_base.json • requirements.txt • architecture.png • project_report.pdf